

151A - MIDTERM

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Problem 1:

a) Here, $p_{n+1} = (p_n)^{\frac{1}{5}} \Rightarrow g(x) = x^{\frac{1}{5}}, [a, b] = \left[\frac{1}{5^{1/4}}, \frac{3}{2}\right]$

To show that $g(x)$ converges to the unique fixed point p such that $g(p) = p$, we need to show that 3 criteria hold true:

① the function $g(x) = x^{\frac{1}{5}}$ is continuous on all of $\mathbb{R} \Rightarrow$ it is continuous on $[a, b]$

$\therefore g \in C[a, b]$ ✓

② $g'(x) = \frac{d}{dx}(x^{\frac{1}{5}}) = \frac{1}{5}x^{-\frac{4}{5}} > 0 \quad \forall x \in \mathbb{R}$

$\therefore g(x)$ is strictly increasing everywhere in $\mathbb{R}$
 $\Rightarrow g(x)$ is strictly increasing on $[a, b]$

Now, $g(a) = g\left(\frac{1}{5^{1/4}}\right) = 5^{-\frac{1}{4}} \approx 0.6687 > \frac{1}{5^{1/4}},$

and $g(b) = g\left(\frac{3}{2}\right) \approx 1.0845 < \frac{3}{2}$

$\therefore$ since g is strictly increasing on $[a, b]$:

$g(x): [a, b] \rightarrow [a, b]$ ✓

③ We saw $g'(x) = \frac{1}{5}x^{-\frac{4}{5}}$. This is only undefined

when $x = 0$. $\therefore g'$ exists on (a, b) .

$g'(x) > 0 \quad \forall x \in \mathbb{R} \Rightarrow |g'(x)| = g'(x)$

$g''(x) = \frac{d}{dx}\left(\frac{1}{5}x^{-\frac{4}{5}}\right) = -\frac{4}{25}x^{-\frac{9}{5}} < 0 \quad \forall x > 0$

$\therefore$ on (a, b) , $g'(x)$ is strictly decreasing.

$g'(a) = g'\left(\frac{1}{5^{1/4}}\right) = 1$

Since $g'(x)$ is strictly decreasing $\Rightarrow$ for all $x \in (a, b)$, $|g'(x)| < 1$... because $|g'(a)| = 1$

$$\therefore |g'(x)| < 1 \quad \forall x \in (a, b) \quad \checkmark$$

Hence, since these 3 criteria are satisfied, by the Convergence of FPMs Theorem, the given sequence $\{p_n\}$ converges to a unique fixed point $p \in [\frac{1}{5^{1/4}}, \frac{3}{2}]$ such that $g(p) = p$.

b) From Part (a), it can be seen easily that $p = 1$ is a fixed-point of $g(x)$: $g(1) = 1^{1/5} = 1 = p$
 $\Rightarrow g(p) = p$

Additionally, we proved that this must be a unique fixed point on the given interval.

$$\therefore \text{as } n \rightarrow \infty, p_n \rightarrow p = 1.$$

If the FPM above were to converge linearly, it would satisfy:

$$\lim_{n \rightarrow \infty} \frac{|p_{n+1} - p|}{|p_n - p|} = \lambda, \quad \lambda \in (0, 1)$$

We have $p_{n+1} = p_n^{1/5}$, and found $p = 1$.

$$\begin{aligned} \therefore \lim_{n \rightarrow \infty} \frac{E_{n+1}}{E_n} &= \lim_{n \rightarrow \infty} \frac{|p_{n+1} - p|}{|p_n - p|} \\ &= \lim_{n \rightarrow \infty} \frac{|p_n^{1/5} - 1|}{|p_n - 1|} \\ &= \lim_{x \rightarrow 1} \frac{|x^{1/5} - 1|}{|x - 1|} \\ &= \left| \lim_{x \rightarrow 1} \frac{x^{1/5} - 1}{x - 1} \right| \end{aligned}$$

This results in $0/0$, so we use L'Hopital's Rule:

$$= \left| \lim_{x \rightarrow 1} \frac{\frac{1}{5} x^{-4/5}}{1} \right|$$

$$= \left| \lim_{n \rightarrow 1} \frac{x^{-4/5}}{5} \right|$$

$$= \left| \frac{1}{5} \right| = \frac{1}{5} = \lambda$$

$$\therefore \frac{E_{n+1}}{E_n} \rightarrow \frac{1}{5} \in (0, 1) \text{ as } n \rightarrow \infty$$

$\Rightarrow p_n$ converges linearly. ■

Problem 2

a) $g(x) = \frac{1}{2} e^{-x+1} (\cos^2(\pi x) + 1)$

$\therefore$ Error ratio:

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{E_{n+1}}{E_n} &= \lim_{n \rightarrow \infty} \left| \frac{P_{n+1} - p}{P_n - p} \right| \\ &= \lim_{n \rightarrow \infty} \left| \frac{g(n) - p}{P_n - p} \right| \\ &= \lim_{x \rightarrow 1} \left| \frac{g(x) - 1}{x - 1} \right| \\ &= \left| \lim_{x \rightarrow 1} \frac{g(x) - 1}{x - 1} \right| \\ &= \left| \lim_{x \rightarrow 1} \frac{\frac{1}{2} e^{-x+1} (\cos^2(\pi x) + 1) - 1}{x - 1} \right| \end{aligned}$$

$\rightarrow$ this gives $\frac{0}{0} \Rightarrow$ use L'Hopital's Rule:

$$\begin{aligned} &= \left| \lim_{x \rightarrow 1} \frac{-\frac{1}{2} e^{-x+1} (\cos^2(\pi x) + 1 + \pi \sin(2\pi x))}{1} \right| \\ &= \left| \lim_{x \rightarrow 1} -\frac{1}{2} e^{-x+1} (\cos^2(\pi x) + 1 + \pi \sin(2\pi x)) \right| \\ &= \left| -\frac{1}{2} \times 2 \right| \\ &= 1 \end{aligned}$$

$\therefore g(x) = \frac{1}{2} e^{-x+1} (\cos^2(\pi x) + 1)$ converges sublinearly.

b) $g(x) = x - \frac{e^{-x+1} (\cos^2(\pi x) + 1) - 2x}{-e^{-x+1} (\cos^2(\pi x) + 1 + \pi \sin(2\pi x)) + 2}$

$$\begin{aligned} \therefore \lim_{n \rightarrow \infty} \frac{E_{n+1}}{E_n} &= \lim_{n \rightarrow \infty} \left| \frac{P_{n+1} - p}{P_n - p} \right| \\ &= \left| \lim_{x \rightarrow 1} \frac{g(x) - 1}{x - 1} \right| \end{aligned}$$

$\rightarrow$ this limit tends to $\frac{0}{0} \Rightarrow$ use L'Hopital's Rule:

$$= \left| \lim_{x \rightarrow 1} \frac{g'(x)}{1} \right|$$

$$= \left| \lim_{x \rightarrow 1} g'(x) \right|$$

$$= \infty$$

$$\therefore g(x) = x - \frac{e^{-x+1}(\cos^2(\pi x) + 1) - 2x}{-e^{-x+1}(\cos^2(\pi x) + 1) + \pi \sin(2\pi x) + 2}$$

diverges.

$$c) \quad g(x) = \frac{19}{20}x + \frac{1}{40}e^{-x+1}(\cos^2(\pi x) + 1)$$

$$\begin{aligned} \therefore \lim_{n \rightarrow \infty} \frac{E_{n+1}}{E_n} &= \lim_{n \rightarrow \infty} \left| \frac{P_{n+1} - P}{P_n - P} \right| \\ &= \left| \lim_{x \rightarrow 1} \frac{g(x) - 1}{x - 1} \right| \\ &= \left| \lim_{x \rightarrow 1} \frac{\frac{19}{20}x + \frac{1}{40}e^{-x+1}(\cos^2(\pi x) + 1) - 1}{x - 1} \right| \end{aligned}$$

$\rightarrow$ tends to $\frac{0}{0} \Rightarrow$ use L'Hopital's Rule:

$$\begin{aligned} &= \left| \lim_{x \rightarrow 1} \frac{\frac{19}{20} + \frac{1}{40}(-e^{-x+1}(\cos^2(\pi x) + 1) + 2\pi \sin(\pi x) \cos(\pi x))}{1} \right| \\ &= \left| \lim_{x \rightarrow 1} \frac{\frac{19}{20} - \frac{1}{40}(e^{-x+1}(\cos^2(\pi x) + 2\pi \sin(\pi x) \cos(\pi x) + 1))}{1} \right| \\ &= \left| \frac{\frac{19}{20} - \frac{2}{40}}{1} \right| = \frac{9}{10} = 0.9 \end{aligned}$$

$$0.9 \in (0, 1)$$

$$\therefore g(x) = \frac{19}{20}x + \frac{1}{40}e^{-x+1}(\cos^2(\pi x) + 1) \quad \text{converges}$$

linearly

$$d) g(x) = 3x - e^{-x+1}(\cos^2(\pi x) + 1)$$

$$\begin{aligned} \therefore \lim_{n \rightarrow \infty} \frac{E_{n+1}}{E_n} &= \lim_{n \rightarrow \infty} \left| \frac{p_{n+1} - p}{p_n - p} \right| \\ &= \left| \lim_{x \rightarrow 1} \frac{g(x) - 1}{x - 1} \right| \\ &= \left| \lim_{x \rightarrow 1} \frac{3x - e^{-x+1}(\cos^2(\pi x) + 1) - 1}{x - 1} \right| \end{aligned}$$

→ this tends to $\frac{0}{0}$ ⇒ use L'Hopital's Rule:

$$\begin{aligned} &= \left| \lim_{x \rightarrow 1} \frac{3 + e^{-x+1}(\cos^2(\pi x) + 2\pi \sin(\pi x)\cos(\pi x) + 1)}{1} \right| \\ &= |3 + 2| = 5 > 1 \end{aligned}$$

$$\therefore g(x) = 3x - e^{-x+1}(\cos^2(\pi x) + 1) \text{ diverges}$$

THEREFORE: my order of preference for choosing functions, arranged in increasing rate of convergence:

$$(b) < (d) < (a) < (c)$$

(does not converge) (does not converge) (sublinear) (linear)

Problem 3

a) $(x_0, y_0) = (-1, 33),$
 $(x_1, y_1) = (0, 51/20),$
 $(x_2, y_2) = (3, -3),$
 $(x_3, y_3) = (2, -21/20),$
 $(x_4, y_4) = (5, 9)$

To find a degree-4 polynomial, we need $L_0(x), L_1(x),$
 $L_2(x), L_3(x)$ and $L_4(x).$

$$L_{n,k}(x) = \prod_{\substack{i=0, \\ i \neq k}}^n \frac{(x-x_i)}{(x_k-x_i)}$$

Hence:

$$\begin{aligned} \bullet L_0(x) &= \frac{(x-x_1)(x-x_2)(x-x_3)(x-x_4)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)(x_0-x_4)} \\ &= \frac{(x-0)(x-3)(x-2)(x-5)}{(-1-0)(-1-3)(-1-2)(-1-5)} \\ &= \frac{1}{72} x(x-3)(x-2)(x-5) \end{aligned}$$

$$\begin{aligned} \bullet L_1(x) &= \frac{(x-x_0)(x-x_2)(x-x_3)(x-x_4)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)(x_1-x_4)} \\ &= \frac{(x+1)(x-3)(x-2)(x-5)}{(0+1)(0-3)(0-2)(0-5)} \\ &= \frac{-1}{30} (x+1)(x-3)(x-2)(x-5) \end{aligned}$$

$$\begin{aligned} \bullet L_2(x) &= \frac{(x-x_0)(x-x_1)(x-x_3)(x-x_4)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)(x_2-x_4)} \\ &= \frac{(x+1)(x-0)(x-2)(x-5)}{(3+1)(3-0)(3-2)(3-5)} \\ &= -\frac{1}{24} x(x+1)(x-2)(x-5) \end{aligned}$$

$$\begin{aligned} \bullet L_3(x) &= \frac{(x-x_0)(x-x_1)(x-x_2)(x-x_4)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)(x_3-x_4)} \\ &= \frac{(x+1)(x-0)(x-3)(x-5)}{(2+1)(2-0)(2-3)(2-5)} \\ &= \frac{1}{18} x(x+1)(x-3)(x-5) \end{aligned}$$

$$\begin{aligned}
 \cdot L_4(x) &= \frac{(x-x_0)(x-x_1)(x-x_2)(x-x_3)}{(x_4-x_0)(x_4-x_1)(x_4-x_2)(x_4-x_3)} \\
 &= \frac{(x+1)(x-0)(x-3)(x-2)}{(5+1)(5-0)(5-3)(5-2)} \\
 &= \frac{1}{180} x(x+1)(x-3)(x-2)
 \end{aligned}$$

$$\text{Now: } P_n(x) = \sum_{k=0}^n f(x_k) \cdot L_{n,k}(x)$$

$$\therefore P_4(x) = y_0 L_0(x) + y_1 L_1(x) + y_2 L_2(x) + y_3 L_3(x) + y_4 L_4(x)$$

$$\begin{aligned}
 \therefore &= \frac{33}{72} x(x-3)(x-2)(x-5) - \frac{51}{600} (x+1)(x-3)(x-2)(x-5) \\
 &+ \frac{3}{24} x(x+1)(x-2)(x-5) - \frac{21}{360} x(x+1)(x-3)(x-5) \\
 &+ \frac{9}{180} x(x+1)(x-3)(x-2)
 \end{aligned}$$

$$\begin{aligned}
 \therefore P_4(x) &= \left(\frac{9}{180} + \frac{11}{25} \right) x^4 + \left(-\frac{9}{45} - \frac{104}{25} \right) x^3 + \left(\frac{9}{180} + \frac{1239}{100} \right) x^2 \\
 &+ \left(\frac{9}{30} - \frac{673}{50} \right) x + \frac{51}{20}
 \end{aligned}$$

Now, we want $\frac{d^3}{dx^3} P_4(x)$ to have a root at $x=1.6$.

$$\begin{aligned}
 \therefore \frac{d}{dx} P_4(x) &= 4 \left(\frac{9}{180} + \frac{11}{25} \right) x^3 + 3 \left(-\frac{9}{45} - \frac{104}{25} \right) x^2 \\
 &+ 2 \left(\frac{9}{180} + \frac{1239}{100} \right) x + \left(\frac{9}{30} - \frac{673}{50} \right)
 \end{aligned}$$

$$\begin{aligned}
 \therefore \frac{d^2}{dx^2} P_4(x) &= 12 \left(\frac{9}{180} + \frac{11}{25} \right) x^2 + 6 \left(-\frac{9}{45} - \frac{104}{25} \right) x \\
 &+ 2 \left(\frac{9}{180} + \frac{1239}{100} \right)
 \end{aligned}$$

$$\therefore \frac{d^3 P_4(x)}{dx^3} = 24 \left(\frac{q}{180} + \frac{11}{25} \right) x + 6 \left(-\frac{q}{45} - \frac{104}{25} \right)$$

Now, to solve for $\frac{d^3 P_4(x)}{dx^3} = 0$:

$$\Rightarrow 24 \left(\frac{q}{180} + \frac{11}{25} \right) x + 6 \left(-\frac{q}{45} - \frac{104}{25} \right) = 0$$

$$\left(\frac{2q}{15} + \frac{264}{25} \right) x = \frac{2q}{15} + \frac{624}{25}$$

$$x = \frac{\frac{2q}{15} + \frac{624}{25}}{\frac{2q}{15} + \frac{264}{25}}$$

$$\Rightarrow x = \frac{5q + 936}{5q + 396} = 1.6 = \frac{8}{5}$$

$$\therefore 25q + 4680 = 40q + 3168$$

$$\therefore 15q = 1512$$

$$\Rightarrow q = \frac{504}{5}$$

$$\therefore P_4(x) = x^4 - \frac{32}{5}x^3 + \frac{259}{20}x^2 - \frac{101}{10}x + \frac{51}{20}$$

b) To find a root for $P_4(x)$, we use Newton's Method.

$$\text{Let } f(x) = P_4(x) = x^4 - \frac{32}{5}x^3 + \frac{259}{20}x^2 - \frac{101}{10}x + \frac{51}{20}$$

The sequence $\{p_n\}_{n=0}^{\infty}$ will converge to the root:

$$p_{n+1} = p_n - \frac{f(p_n)}{f'(p_n)}$$

We have $p_0 = 1.213$. Using MATLAB to perform iterations for p_n until $n = 100$, we get:

$$p_{100} = 1.500000$$

Problem 4

$$a) \quad p_{n+1} = \frac{1}{2} \left(p_n + \frac{\alpha}{p_n} \right) \quad \Rightarrow \quad g(x) = \frac{x}{2} + \frac{\alpha}{2x}$$

$g(x)$ is undefined when $x=0 \Rightarrow g \in \mathbb{C}$ when $x > 0$ ✓

If we choose $p_0 > 0$, and $\alpha > 0$, then $\frac{1}{2} \left(p_n + \frac{\alpha}{p_n} \right) > 0$.

$$\therefore g: (0, \infty) \mapsto (0, \infty) \quad \checkmark$$

$$\begin{aligned} g'(x) &= \frac{d}{dx} \left(\frac{1}{2} \left(x + \frac{\alpha}{x} \right) \right) = \frac{1}{2} - \frac{\alpha}{2x^2} \\ &= \frac{x^2 - \alpha}{2x^2} \end{aligned}$$

$$\Rightarrow g'(x) = 0 \quad \text{when} \quad x = \sqrt{\alpha} \quad (> 0)$$

$$\begin{aligned} \text{when } x \in (0, \sqrt{\alpha}), \quad g'(x) &< 0 \\ \text{when } x \in (\sqrt{\alpha}, \infty), \quad g'(x) &> 0 \end{aligned}$$

$$\text{However, for } g'(x) = 1: \quad \frac{x^2 - \alpha}{2x^2} = 1$$

$$\Rightarrow x^2 - \alpha = 2x^2$$

$$\Rightarrow x^2 = -\alpha$$

this is not possible if $\alpha > 0$

$\therefore g'(x)$ never equals 1 for $x > 0$.

$$\therefore \forall x > 0, \quad g'(x) < 1$$

Hence:

$$\begin{aligned} \checkmark \quad &g(x) \text{ is continuous for } x > 0 \\ \checkmark \quad &g(x) > 0 \text{ when } x > 0 \\ \checkmark \quad &g'(x) < 1 \text{ when } x > 0 \end{aligned}$$

Thus, $g(x)$ converges to a fixed point $p \in (0, \infty)$,
when $p_0 > 0$.

$$b) \quad g(x) = \frac{1}{2} \left(x + \frac{\alpha}{x} \right) = \frac{1}{2} \left(\frac{x^2 + \alpha}{x} \right)$$

If $x < 0$, then $x^2 > 0$, but $\frac{x^2 + \alpha}{x} < 0$

$\therefore$ if p_0 , then $p_n < 0 \quad \forall n \geq 0$

Hence, we want to consider the interval $x \in (-\infty, 0)$.

$g(x)$ is undefined when $x = 0 \Rightarrow g \in \mathbb{C}$ when $x < 0$ ✓

As shown above, if $x \in (-\infty, 0)$, then $g(x) \in (-\infty, 0)$

$\therefore g: (-\infty, 0) \rightarrow (-\infty, 0)$ ✓

$$\begin{aligned} g'(x) &= \frac{1}{2} - \frac{\alpha}{2x^2} \\ &= \frac{x^2 - \alpha}{2x^2} \Rightarrow g'(x) = 0 \text{ when } x = -\sqrt{\alpha} \quad (x < 0) \end{aligned}$$

when $x \in (-\sqrt{\alpha}, 0)$, $g'(x) < 0$

when $x \in (-\infty, -\sqrt{\alpha})$, $g'(x) > 0$

However, for $g'(x) = 1$: $\frac{x^2 - \alpha}{2x^2} = 1$

$$\Rightarrow x^2 - \alpha = 2x^2$$

$$\Rightarrow x^2 = -\alpha$$

this is not possible if $\alpha > 0$

$\therefore g'(x)$ never equals 1 for $x < 0$.

$\therefore \forall x < 0, \quad g'(x) < 1$

Hence:

- ✓ $g(x)$ is continuous for $x < 0$
- ✓ $g(x) < 0$ when $x < 0$
- ✓ $g'(x) < 1$ when $x < 0$

Thus, $g(x)$ converges to a fixed point $p \in (-\infty, 0)$ when $p_0 < 0$.

Additionally, if $p_0 > 0$ converges to some value p , then when $p_0 < 0$, p_n will converge to $-p$.

To prove this, show that $g(x)$ is an odd function:

$$\begin{aligned} g(-x) &= \frac{1}{2} \left(-x + \frac{\alpha}{-x} \right) \\ &= \frac{1}{2} \left(\frac{-x^2 + \alpha}{-x} \right) \\ &= -\frac{1}{2} \left(\frac{x^2 + \alpha}{x} \right) = -g(x) \end{aligned}$$

Hence, if $g(p) = p$, then $g(-p) = -g(p) = -p$
 $\Rightarrow -p$ is a fixed point

$\therefore$ if $p_0 > 0$, $p_n \rightarrow p$ (fixed point), and
if $p_0 < 0$, $p_n \rightarrow -p$ (fixed point).